

# EXCITATION-DEPENDENT IDENTIFIABILITY OF LATENT HIGHER-ORDER STRUCTURE FROM EDGE FLOWS

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## ABSTRACT

Which triangles of a network carry higher-order interactions? Edge flows reveal the latent filled-triangle set  $\mathcal{S}$  only through their curl, and we show that *what* is identifiable is governed by the excitation covariance  $\Gamma_{\mathcal{S}}$  of the triangle signals, through three regimes. For positive-diagonal  $\Gamma_{\mathcal{S}}$  (known or unknown), the weighted support is identifiable at *any* rank deficiency of  $\mathbf{B}_2$ , because the lifted signatures  $\{\mathbf{b}_\tau \mathbf{b}_\tau^\top\}$  are always linearly independent. For unrestricted PSD  $\Gamma_{\mathcal{S}}$ , the population curl covariance  $\Sigma_z$  determines only the achievable set  $\mathcal{C}_{\mathcal{S}} = \{\mathbf{M} \succeq 0 : \text{im } \mathbf{M} \subseteq \text{im } \mathbf{U}_{\mathcal{S}}\}$ , hence the support solely up to  $\{\mathcal{S} : \text{im } \mathbf{M} \subseteq \text{im } \mathbf{U}_{\mathcal{S}}\}$ ; a strictly positive-diagonal but correlated  $\Gamma_{\mathcal{S}}$  already defeats identification (a rank-one witness on  $K_4$ ). At the projector excitation  $\Gamma_{\mathcal{S}} = \sigma_c^2 (\mathbf{B}_{2,\mathcal{S}}^\top \mathbf{B}_{2,\mathcal{S}})^+$  the covariance collapses to  $\sigma_n^2 \mathbf{I} + \sigma_c^2 \mathbf{P}_{\text{im}}$  exactly, so equal-image supports are indistinguishable at every SNR and sample size (noise level known throughout). This projector is exactly the Hodge-smoothness prior used in prior topology learning [1]. In the diagonal regime we give a lifted-covariance non-negative least-squares estimator that is consistent with a fully explicit  $O(1/N)$  failure bound, validated on  $K_4$ – $K_8$  against subspace and greedy baselines, including an excitation interpolation under which equal-image distinguishability provably vanishes. On real data, the same curl statistic *localizes* genuine unplanted rotational structure — tropical cyclones in ERA5 winds — against a same-field vorticity reference and the independent IB-TrACS archive. Code, data, and a 49-test suite numerically checking every constant are released.

**Index Terms**— Topological signal processing, simplicial complexes, identifiability, Hodge decomposition, covariance estimation, edge flows.

## 1. INTRODUCTION

Edge flows — currency exchange, traffic, ranking comparisons — carry higher-order structure on the triangles of a simplicial complex [2, 3, 4]. Topological signal processing builds filters and detectors on the Hodge Laplacian [5, 6, 7, 8, 9], and a recurring primitive is that the set  $\mathcal{S}$  of “filled” triangles is unknown. One line of work *estimates*  $\mathcal{S}$  algorithmically: MAP estimation under a Hodge-smoothness prior [1], greedy topology learning [10], sparse cell-complex augmentation [11]; second-order precedent is older still — Barbarossa and Sardellitti infer filled triangles from the sample covariance of edge flows by a PCA/total-variation criterion [2, §VII]. A second line *detects* whether a signal lies in a *known* Hodge subspace: matched topological subspace detectors [12]. All of these fix a generative or observation model and give an algorithm; none states *when* the latent

structure is identifiable. That is the gap: *when is the latent structure that generates the flows identifiable at all — and what part of it?*

The answer is not a property of the geometry alone: it depends on the *excitation*. We model triangle signals as  $\mathbf{y}_{\mathcal{S}} \sim \mathcal{N}(\mathbf{0}, \Gamma_{\mathcal{S}})$  with PSD excitation covariance  $\Gamma_{\mathcal{S}}$  and show that the identifiable object moves with the regime of  $\Gamma_{\mathcal{S}}$  (Theorem 1): from the full weighted support (positive diagonal, any rank deficiency), to only the achievable-covariance set  $\mathcal{C}_{\mathcal{S}}$  — so the support merely up to  $\{\mathcal{S} : \text{im } \mathbf{M} \subseteq \text{im } \mathbf{U}_{\mathcal{S}}\}$  — under unrestricted PSD excitation, to provable indistinguishability of equal-image supports at the projector excitation. The last case is not a corner curiosity: the Hodge-smoothness prior of [1],  $\mathbf{y}_{\mathcal{S}} \sim \mathcal{N}(\mathbf{0}, \mathbf{L}_2^+)$  with  $\mathbf{L}_2 = \mathbf{B}_2^\top \mathbf{B}_2$ , is this projector excitation — so their model is second-order-unidentifiable and succeeds only through its sparsity prior. Our result thus both corrects a folklore “rank obstruction” and delimits what covariance- and image-level methods [2, 12] can resolve. Our tools are classical — detection theory and Fano arguments [13, 14], in the lineage of support-recovery and sparse-covariance limits [15, 16, 17, 18] — applied through a reduction (curl annihilation, after HodgeRank’s curl-as-inconsistency reading [19]) that makes them bite. Contributions:

1. *Theory*: the three excitation regimes (Thm. 1), with global analytic separation constants for the isotropic regime — equal-image binary supports are separated by at least 9 (attained by adding a tetrahedron’s fourth face) and equal-cardinality supports by 16 (swaps), proved for *every* clique complex via a Johnson-graph eigenvalue argument — and Chernoff exponents, for fixed graph and support as  $\rho_2 \rightarrow 0$ , whose worst case equals one isolated-triangle detection (Thm. 2).
2. *Achievability*: a lifted-covariance NNLS estimator for the diagonal class, consistent with a fully explicit  $O(1/N)$  failure bound in which every constant is derived and numerically checked (Thm. 3); Monte-Carlo validation on  $K_4$ – $K_8$  with random supports against oracle-aided subspace and greedy baselines, and an excitation-interpolation experiment in which equal-image distinguishability collapses as predicted (Fig. 1).
3. *Real data, carefully positioned*: curl-based *vortex localization* — not topology recovery — of unplanted tropical-cyclone circulation in ERA5 winds, validated against a same-field vorticity reference and the independent IB-TrACS archive, with a classical baseline and moving-block bootstrap confidence intervals (Fig. 2).

## 2. MODEL AND CURL REDUCTION

Fix a graph  $(\mathcal{V}, \mathcal{E})$  with node–edge incidence  $\mathbf{B}_1$  and, for a candidate triangle set  $\mathcal{T}$  (3-cliques), edge–triangle incidence  $\mathbf{B}_2 \in \mathbb{R}^{|\mathcal{E}| \times p}$ ;

$B_1 B_2 = \mathbf{0}$ , and each column  $\mathbf{b}_\tau$  has three  $\pm 1$  entries, so  $\mathbf{G} = B_2^\top B_2$  has  $G_{\tau\tau} = 3$  and  $G_{\sigma\tau} \in \{0, \pm 1\}$  (nonzero iff the triangles share an edge). We observe  $N$  i.i.d. snapshots

$$\mathbf{f}_t = B_1^\top \mathbf{a}_t + B_{2,S} \mathbf{y}_t + \mathbf{h}_t + \mathbf{n}_t, \quad t = 1, \dots, N, \quad (1)$$

with gradient nuisance  $\mathbf{a}_t \sim \mathcal{N}(\mathbf{0}, \sigma_g^2 \mathbf{I})$ , harmonic nuisance  $\mathbf{h}_t$ , noise  $\mathbf{n}_t \sim \mathcal{N}(\mathbf{0}, \sigma_n^2 \mathbf{I})$ , and triangle excitation

$$\mathbf{y}_t \sim \mathcal{N}(\mathbf{0}, \Gamma_S), \quad \Gamma_S \succeq \mathbf{0}, \quad (2)$$

where  $\Gamma_S \succeq \mathbf{0}$  is the excitation covariance; making it explicit is the point of this paper (prior generative models are special cases of it, §1). We take the harmonic nuisance to be *candidate-orthogonal*,  $\mathbf{h}_t \in \ker B_2^\top$  (orthogonal to the curl space of the *full* candidate complex); the gradient part is annihilated unconditionally since  $B_2^\top B_1^\top = \mathbf{0}$ . Then curl projection removes both nuisances exactly [19]: in an orthonormal basis  $\mathbf{Q}$  of  $\text{im } B_2$  ( $r = \text{rank } B_2$ ,  $B_2$  the full candidate incidence), the sufficient statistic  $\mathbf{z}_t = \mathbf{Q}^\top \mathbf{f}_t$  is Gaussian with

$$\Sigma_z(\mathcal{S}, \Gamma_S) = \sigma_n^2 \mathbf{I}_r + U_S \Gamma_S U_S^\top, \quad U = \mathbf{Q}^\top B_2, \quad (3)$$

with  $\|\mathbf{u}_\tau\|^2 = 3$  and  $\mathbf{u}_\sigma^\top \mathbf{u}_\tau = G_{\sigma\tau}$ . Identifiability of  $\mathcal{S}$  from (3) is the object of study; for the isolated-triangle scalar test this reduces to the classical two-variance problem with curl-SNR  $\rho = 3\sigma_c^2/\sigma_n^2$  [13] (supplement §S1–S3 develops that line: detection thresholds, Fano converses, decorrelated detectors, plug-in estimation).

**Definition 1** (Observational equivalence and identifiability). *All statements are about the population covariance  $\Sigma_z$  (the  $N \rightarrow \infty$  limit of  $\hat{\Sigma}_z$ ); finite- $N$  estimation is §4. Two pairs  $(\mathcal{S}, \Gamma_S)$  and  $(\mathcal{S}', \Gamma_{S'})$  are observationally equivalent if  $\Sigma_z(\mathcal{S}, \Gamma_S) = \Sigma_z(\mathcal{S}', \Gamma_{S'})$ ; a functional of  $(\mathcal{S}, \Gamma_S)$  is identifiable on an excitation class  $\mathcal{G}$  if it is constant on each equivalence class with excitations in  $\mathcal{G}$ . The realized range is  $\mathcal{R}(\mathcal{S}, \Gamma_S) = \text{im}(\Sigma_z - \sigma_n^2 \mathbf{I}) = \text{im}(U_S \Gamma_S U_S^\top) \subseteq \text{im } U_S$ , always identifiable since it is a function of  $\Sigma_z$  alone. If  $\Gamma_S \succ \mathbf{0}$  then  $\mathcal{R} = \text{im } U_S$ ; when  $B_{2,S}$  has full column rank (so  $U_S$  is injective) the converse holds as well.*

### 3. EXCITATION-DEPENDENT IDENTIFIABILITY

**Lemma 1** (Lifted spark). *For any candidate set of distinct 3-cliques of a simple graph, the matrices  $\{\mathbf{b}_\tau \mathbf{b}_\tau^\top\}_{\tau \in \mathcal{T}}$  are linearly independent.*

*Proof.* A pair of distinct edges lies in at most one common triangle, so for  $e \neq e'$  both in  $\tau$ , the  $(e, e')$  entry of  $\sum_\sigma w_\sigma \mathbf{b}_\sigma \mathbf{b}_\sigma^\top$  equals  $\pm w_\tau$ : every coefficient is read off a distinct matrix entry.  $\square$

**Theorem 1** (Three excitation regimes). *Let  $\sigma_n^2$  be known and write  $\mathbf{M} = \mathbf{M}(\mathcal{S}, \Gamma) = U_S \Gamma U_S^\top = \Sigma_z - \sigma_n^2 \mathbf{I}$  in (3). The regimes below are overlapping constraints on  $\Gamma_S$ , not a partition (e.g.  $\sigma_c^2 \mathbf{I}$  is both diagonal (a) and a special full-rank PSD (b)). (a) Positive diagonal. If  $\Gamma_S = \text{diag}(\gamma_\tau)_{\tau \in \mathcal{S}}$  with  $\gamma_\tau > 0$  (known or unknown), then  $\mathbf{M} = \sum_{\tau \in \mathcal{S}} \gamma_\tau \mathbf{u}_\tau \mathbf{u}_\tau^\top$  and, by Lemma 1, the map  $(\mathcal{S}, \Gamma_S) \mapsto \Sigma_z$  is injective: the weighted support is identifiable at any rank deficiency of  $B_2$ , including all  $\binom{n}{3}$  triangles of  $K_n$ . (b) Arbitrary PSD. The achievable set  $\mathcal{C}_S = \{U_S \Gamma U_S^\top : \Gamma \succeq \mathbf{0}\} = \{\mathbf{M} \succeq \mathbf{0} : \text{im } \mathbf{M} \subseteq \text{im } U_S\}$ , so the population covariance determines  $\mathbf{M}$  in full but pins the support only to  $\{\mathcal{S} : \text{im } \mathbf{M} \subseteq \text{im } U_S\}$ : the realized range  $\mathcal{R} = \text{im } \mathbf{M} \subseteq \text{im } U_S$  is always readable, yet the candidate image  $\text{im } B_{2,S}$  is not (a larger-image  $\mathcal{S}'$  with a suitable  $\Gamma$  gives the same*

$\mathbf{M}$ ;  $\Gamma_S \succ \mathbf{0}$  does recover  $\mathcal{R} = \text{im } U_S$ ). The obstruction is correlation, not zero variance: on  $K_4$  (all four faces) with  $\mathbf{c} \in \ker U_S$  and  $\mathbf{v} = \mathbf{e}_1 - \frac{1}{4}\mathbf{c} = (.75, .25, -.25, .25)^\top$ , the strictly-positive-diagonal  $\Gamma' = \mathbf{v}\mathbf{v}^\top$  gives  $U_S \Gamma' U_S^\top = \mathbf{u}_1 \mathbf{u}_1^\top$  — identical to the singleton  $\{\tau_1\}$  though the images are 3- vs. 1-dimensional. Regime (a) additionally requires  $\Gamma$  diagonal. (c) Projector excitation. For  $\Gamma_S = \sigma_c^2 (B_{2,S}^\top B_{2,S})^+$ ,  $\mathbf{M} = \sigma_c^2 \mathbf{P}_{\text{im } U_S}$  exactly; supports with equal curl image induce identical snapshot distributions and are indistinguishable at every SNR and every  $N$ .

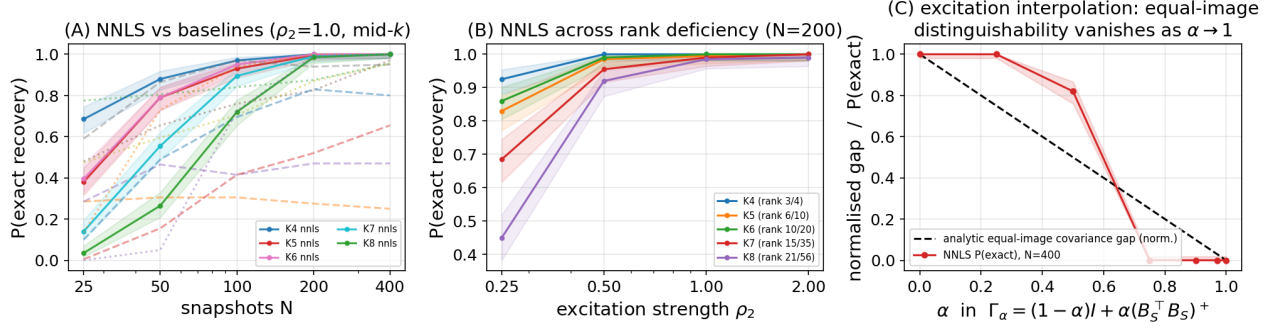
*Proof sketch.* (a) Read the weights off the lifted representation (spark). (b)  $\subseteq$  is immediate;  $\supseteq$ : given PSD  $\mathbf{M}$  with  $\text{im } \mathbf{M} \subseteq \text{im } U_S$ , take  $\Gamma = U_S^\top \mathbf{M} (U_S^\top)^\top \succeq \mathbf{0}$ . (c)  $\mathbf{A}(\mathbf{A}^\top \mathbf{A})^+ \mathbf{A}^\top = \mathbf{P}_{\text{im } \mathbf{A}}$  (SVD). On  $K_n$ ,  $\dim \text{im } B_2 = \binom{n-1}{2}$  (a  $3/n$  DoF ratio), and equal-image supports exist whenever a tetrahedron is hosted ( $\mathbf{b}_{012} - \mathbf{b}_{013} + \mathbf{b}_{023} - \mathbf{b}_{123} = \mathbf{0}$ ). Full proofs and numerical certificates: supplement §S4, tests/test\_excitation.py.  $\square$

**Remark 1** (Scope). (i) *Deterministic unknown signals are the rank-one degenerate boundary of regime (b): the realized range is a single direction, so not even the image is seen.* (ii) *On  $K_n$  with  $\sigma_n$  unknown, regime (a) retains exactly one ambiguous direction, since  $\sum_{\tau \in \mathcal{T}} \mathbf{u}_\tau \mathbf{u}_\tau^\top = n \mathbf{I}_r$ : a uniform weight shift trades off against the noise floor.* (iii) *Regime (b) delimits image-level methods — e.g. matched subspace detectors [12] — whose population statistics see only the realized range  $\text{im } \mathbf{M} \subseteq \text{im } B_{2,S}$ : without excitation structure, no method can do better.*

**Theorem 2** (The price in the isotropic class). *Let  $\Gamma_S = \sigma_c^2 \mathbf{I}$ ,  $\rho_2 = \sigma_c^2/\sigma_n^2$ , and write the dimensionless signature Gram  $\mathbf{D}_S = \sum_{\tau \in \mathcal{S}} \mathbf{u}_\tau \mathbf{u}_\tau^\top$  (so  $\mathbf{M}_S = \sigma_c^2 \mathbf{D}_S$ ). For any clique complex: distinct binary supports with equal curl image satisfy  $\|\mathbf{D}_S - \mathbf{D}_{S'}\|_F^2 \geq 9$ , with equality exactly for single-face differences (e.g.  $\mathcal{S}' = \mathcal{S} \cup \{\text{fourth face}\}$ ), and  $\geq 16$  under equal cardinality (tetrahedral swaps). For fixed graph and support pair, as  $\rho_2 \rightarrow 0$  the optimal-test Chernoff exponents are  $C_G = \rho_2^2 \|\Delta \mathbf{D}\|_F^2 / 16 (1 + o(1))$ , i.e.  $\frac{9}{16} \rho_2^2 (1 + o(1))$  (subset) and  $\rho_2^2 (1 + o(1))$  (swap) — and  $\frac{9}{16} \rho_2^2$  equals the isolated-triangle detection exponent  $C(\rho) = \rho^2/16$  at  $\rho = 3\rho_2$ : the hardest equal-image decision costs one isolated-triangle detection, so rank deficiency is free at the exponent level and only the Fano log-multiplicity of the  $4\binom{n}{4}$  tetrad hypotheses ( $\asymp \log n$  on  $K_n$ ) reflects the geometry.*

*Proof sketch.* Separations:  $\|\sum_\tau c_\tau \mathbf{b}_\tau \mathbf{b}_\tau^\top\|_F^2 = \mathbf{c}^\top (9\mathbf{I} + \mathbf{A}) \mathbf{c}$  for ternary  $\mathbf{c}$ , where  $\mathbf{A}$  is the share-an-edge adjacency — an induced subgraph of the Johnson graph  $J(n, 3)$ , so  $\lambda_{\min}(\mathbf{A}) \geq -3$  by interlacing and  $\text{sep} \geq 6\|\mathbf{c}\|^2$ ; enumerate  $\|\mathbf{c}\|^2 \in \{1, 2\}$ . Exponents: Bhattacharyya expansion gives  $C_G = \rho_2^2 \|\Delta \mathbf{D}\|_F^2 / 16 (1 + o(1)) = \|\Delta \mathbf{M}\|_F^2 / (16\sigma_n^4) (1 + o(1))$  (the two forms coincide). Numerically checked: swap  $C_G/\rho_2^2$  within  $6 \times 10^{-4}$  of 1 and subset matching the single-triangle exponent to  $3 \times 10^{-5}$  relative (at  $\rho_2 = 10^{-4}, 10^{-5}$ ); separations exhaustive over all  $2^{10}$  supports of  $K_5$ . Proofs: supplement §S4.  $\square$

**Proposition 1** (Edge-sharing marginal laws). *Assume  $B_2$  full column rank and  $\Gamma_S = \sigma_c^2 \mathbf{I}$ . The GLS/BUE inversion  $\hat{\mathbf{y}}_t = \mathbf{G}^{-1} B_2^\top \mathbf{f}_t$  [20] satisfies, exactly,  $\hat{\mathbf{y}}_t = \mathbf{y}_t + \mathbf{e}_t$  with  $\mathbf{e}_t \sim \mathcal{N}(\mathbf{0}, \sigma_n^2 \mathbf{G}^{-1})$ : each coordinate is a two-variance test with effective SNR  $\rho_\tau^{\text{eff}} = \sigma_c^2 / (\sigma_n^2 (\mathbf{G}^{-1})_{\tau\tau})$  — the classical variance-inflation factor as the price of edge-sharing — and exact recovery admits a correlation-robust union-bound guarantee. The noise coordinates stay correlated ( $\sigma_n^2 \mathbf{G}^{-1}$  is not diagonal), so the joint law does not factorize; benchmarks, the naive-detector failure mode, and proofs are in supplement §S1, S3.*



**Fig. 1.** Second-order achievability. (A) Exact recovery vs.  $N$  on  $K_4$ – $K_8$  (random supports,  $\rho_2 = 1$ ): NNLS (solid, 95% Wilson bands) vs. oracle-aided subspace (dashed) and greedy (dotted) baselines. (B) NNLS vs.  $\rho_2$  at  $N = 200$  across rank deficiency (legend: rank  $\mathbf{B}_2/p$ ). (C) Excitation interpolation on the  $K_5$  tetrad: the analytic equal-image gap (normalized) closes only at  $\alpha = 1$  (Thm. 1(c)); thresholded NNLS recovery collapses earlier (zero from  $\alpha = 0.75$ ).

**Proposition 2** (Detection and joint-recovery scalings). *In the isotropic class, deciding one isolated triangle is the two-variance Gaussian test with variances  $3\sigma_n^2$  vs.  $9\sigma_c^2 + 3\sigma_n^2$  [13]: attaining error  $\leq \delta$  at the exponent level requires the curl-SNR scale  $\rho \geq \rho^*(N) = \Theta(\sqrt{\log(1/\delta)/N})$  ( $C(\rho) = \rho^2/16 + o(\rho^2)$  [14]). Recovering a uniformly random size- $k$  support among  $p$  edge-disjoint candidates with error  $\leq \epsilon$  requires  $N \geq [(1-\epsilon)\log(\binom{p}{k}) - \log 2]/(k \text{KL}_1)$ ,  $\text{KL}_1 = (\rho - \log(1+\rho))/2$  (Fano); joint recovery pays a  $\log(\binom{p}{k})$  factor. Signal-agnostic and sub-Gaussian variants, the exact edge-disjoint product law, partial edge observation ( $q^{3k}$  laws), and plug-in calibration are developed in supplement §S1–S3; these scalings are classical [15] — what is problem-specific is the reduction that produces  $\rho$  and the constants (3, 9).*

#### 4. ACHIEVABILITY: LIFTED-COVARIANCE NNLS

In class (a) the theory says the weighted support is identifiable; the matching estimator is non-negative least squares on the lifted dictionary  $\mathbf{A} = [\text{vec}(\mathbf{u}_\tau \mathbf{u}_\tau^\top)]_\tau$ :

$$\hat{\mathbf{w}} = \arg \min_{\mathbf{w} \geq 0} \left\| \hat{\Sigma}_z - \sigma_n^2 \mathbf{I} - \sum_\tau w_\tau \mathbf{u}_\tau \mathbf{u}_\tau^\top \right\|_F^2, \quad (4)$$

with  $\hat{\Sigma}_z = \frac{1}{N} \sum_t \mathbf{z}_t \mathbf{z}_t^\top$ ,  $\hat{\mathcal{S}} = \{\tau : \hat{w}_\tau > w_{\min}/2\}$ , and  $w_{\min} = \min_{\tau \in \mathcal{S}} w_\tau$ .

**Theorem 3** (Consistency and explicit failure bound).  *$\hat{\mathbf{w}} \rightarrow \mathbf{w}$  almost surely, and for every  $N$ ,*

$$\Pr[\hat{\mathcal{S}} \neq \mathcal{S}] \leq \frac{16[(\text{tr} \Sigma_z)^2 + \|\Sigma_z\|_F^2]}{N \sigma_{\min}^2(\mathbf{A}) w_{\min}^2}. \quad (5)$$

*Proof sketch.* Three derived steps, each numerically checked: (i) cone-constrained least squares obeys the deterministic bound  $\|\hat{\mathbf{w}} - \mathbf{w}\|_2 \leq 2\|\hat{\Sigma}_z - \Sigma_z\|_F / \sigma_{\min}(\mathbf{A})$  — optimality of  $\hat{\mathbf{w}}$ , feasibility of  $\mathbf{w}$ , Cauchy–Schwarz — with  $\sigma_{\min}(\mathbf{A}) > 0$  by Lemma 1; (ii) for Gaussian snapshots  $\mathbb{E}\|\hat{\Sigma}_z - \Sigma_z\|_F^2 = [(\text{tr} \Sigma_z)^2 + \|\Sigma_z\|_F^2]/N$  exactly (Wishart second moments); (iii) Markov at threshold  $w_{\min}/2$ . The bound is conservative (Markov) — its value is the explicit  $O(1/N)$  rate with computable constants; empirically the transition occurs roughly an order of magnitude earlier (Fig. 1). Proof: supplement §S4.  $\square$

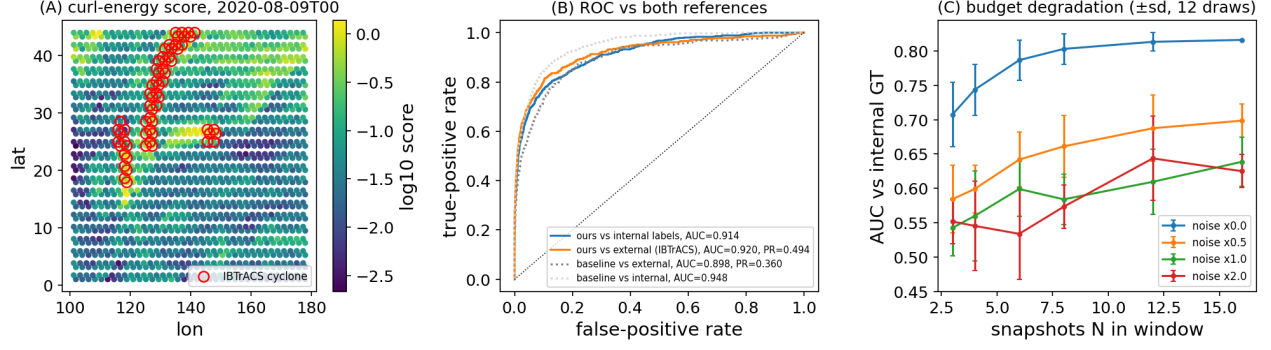
Fig. 1 validates the estimator across  $K_4$ – $K_8$  with per-trial random supports at several sparsities  $k$ , a full  $(N, \rho_2)$  grid with

$N \in \{25, \dots, 400\}$  and  $\rho_2 \in \{0.25, \dots, 2\}$ , 200 Monte-Carlo trials per cell, and 95% Wilson intervals (run\_second\_order.py; every cell, including the derived bound (5) evaluated at that cell, is in the released JSON). Protocol notes: the headline threshold is the theorem’s rule  $w_{\min}/2$  (planted weights known by design); an oracle-free largest-gap rule is run alongside and logged per cell — it is worse than the theorem’s rule (grid-mean exact-recovery 0.63 vs. 0.69, ahead in only 18 of 260 cells), reported for completeness, not as an improvement. The full edge-domain pipeline is used, so the gradient nuisance is present and annihilated by the projection; the greedy comparator is a generic covariance-atom heuristic (not a tuned literature method), and both baselines receive oracle side information — the subspace method the true  $k$  and the true  $\dim \mathbf{B}_{2,S}$ , the greedy fitter the true  $\sigma_n$ .

(A) NNLS is the only estimator attaining exact recovery in every regime — including maximal rank deficiency, e.g. rank  $\mathbf{B}_2/p = 21/56$  on  $K_8$  — reaching probability 1 at the plotted mid- $k$  cells by  $N=400$ ,  $\rho_2=1$  (0.99 at the densest  $K_8$  cell,  $k=28$ ). The oracle- $k$  subspace baseline can lead at small  $N$  but collapses on dependent candidates (0.00 at  $K_8$ ,  $k=28$ ): its population scores tie there (Remark 1(iii)), and its finite- $N$  tie-breaking rides on sample eigen-anisotropy, a side channel the projector excitation removes entirely. Greedy stalls at sparse supports (0.43 on  $K_5$  at  $k=2$ ,  $N=400$ ). (B) Recovery improves monotonically in  $\rho_2$  uniformly across rank deficiency — the empirical face of Theorem 1(a). The bound (5) upper-bounds every cell; being a Markov bound it is conservative by one to three orders of magnitude (the empirical failure decays exponentially, the bound as  $1/N$ ), and it is certified non-vacuously at every informative cell — down to 0.42 on the released grid,  $\sim 0.75$  at the dedicated test cells, with zero observed failures at all of them. (C) Under  $\Gamma_\alpha = \rho_2[(1-\alpha)\mathbf{I} + \alpha(\mathbf{B}_{2,S}^\top \mathbf{B}_{2,S})^+]$  on the  $K_5$  tetrad, with the same  $w_{\min}/2$  rule (the diagonal model is deliberately misspecified for  $\alpha > 0$  — that is the point), the analytic equal-image gap shrinks continuously and the empirical NNLS recovery of the specific  $\mathcal{S}$  collapses before the gap reaches zero — thresholded estimation dies earlier than distinguishability — with the threshold-free endpoint at  $\alpha = 1$  exactly on Theorem 1(c): identical covariances, no method can identify  $\mathcal{S}$ . Identifiability is a property of the excitation, not the geometry alone.

#### 5. REAL DATA: CURL-BASED VORTEX LOCALIZATION

We position the real-data study precisely: it is *vortex localization*, not filled-triangle topology recovery — the mesh below is



**Fig. 2.** Curl-based vortex localization on real data. (A) Centered curl-energy scores (vorticity scale) for one 4-day window; red circles: independent IBTrACS fixes. (B) ROC pooled over all windows: same-field vorticity reference, independent IBTrACS labels (with PR-AUC), and the classical coarse-vorticity baseline. (C) Budget degradation (mean  $\pm$  sd over 12 subsample draws; uniform centered statistic,  $N \geq 3$ ).

simply connected, hence full-column-rank with no equal-image confusers, and real weather carries no latent boolean support. What it demonstrates is that the paper’s sufficient statistic localizes *genuine, unplanted* rotational structure on a real vector field. Construction: ERA5 10m winds [21] over the Western North Pacific (Aug–Sep 2020,  $0.7^\circ/6$ -hourly) are subsampled to a  $2.1^\circ$  right-triangulated mesh (836 nodes, 2389 edges, 1554 triangles;  $\# \text{triangles} = |\mathcal{E}| - |\mathcal{V}| + 1$  by Euler, so  $\mathbf{B}_2$  is full column rank); edge flows are trapezoidal line integrals of the wind in a local equirectangular metric. By Stokes’ theorem a triangle’s curl is its circulation, so the temporally centered curl-energy score (area<sup>2</sup>-normalized to the mean-vorticity scale; no oracle parameters) ranks cyclone-bearing triangles within 4-day windows of 16 snapshots, pooled over the season’s 15 windows for evaluation. Storms translate several triangle-widths per 4-day window; centering removes the window-mean flow, so the statistic responds to the time-varying circulation a moving vortex deposits. Validation (Fig. 2B) against a *same-field consistency reference* (finite-difference vorticity on the  $3\times$ -finer source grid) gives AUC 0.914 (moving-block bootstrap 95% CI [0.894, 0.936]), and against the *genuinely independent* IBTrACS archive [22] — 13 storms, 726 fixes — AUC 0.920 [0.879, 0.951], PR-AUC 0.494 [0.273, 0.666] at prevalence 1.8%, and  $P@k = 0.483$ . Against a classical pointwise-vorticity baseline at the same information budget (mesh-node winds), the edge-flow statistic is better on *all three* independent external metrics (baseline: AUC 0.898, PR-AUC 0.360,  $P@k = 0.389$ ) and slightly lower on the internal reference (0.914 vs. 0.948), which shares the baseline’s functional. The decorrelated (GLS) variant of Prop. 1 ranks worse here ( $\mathbf{G}^{-1}$  amplifies noise on a near-regular mesh) and is reported in the repository. FX consistency checks and planted road-network studies from earlier revisions are retained in the repository and supplement, not claimed as recovery evidence.

**Limitations.** The i.i.d.-Gaussian snapshot model is an idealization of windowed weather; evaluation is therefore threshold-free ranking rather than support recovery, moving-block bootstrap intervals quantify pooling uncertainty, and the internal reference is a same-field consistency check — only IBTrACS is independent. The vorticity-threshold sensitivity of the internal AUC (0.876–0.959 over  $10^{-5}$ – $5 \times 10^{-5} \text{ s}^{-1}$ ) is reported in the released JSON. Nothing here claims a new cyclone detector; the experiment grounds the paper’s sufficient statistic on genuinely unplanted structure.

## 6. CONCLUSION

Identifiability of latent higher-order structure from edge flows is excitation-dependent: fully identifiable under diagonal excitation at any rank deficiency; under unrestricted PSD excitation the population covariance pins down only the achievable-covariance set  $\mathcal{C}_S$ , so the support is determined merely up to  $\{S : \text{im } \mathbf{M} \subseteq \text{im } \mathbf{U}_S\}$  (not even the candidate image in general); and provably ambiguous at the projector excitation — with global analytic separation constants, a worst-case exponent equal to one isolated-triangle detection, and a consistent NNLS estimator carrying a fully explicit  $O(1/N)$  bound. The three regimes delimit both algorithmic structure learning [1, 10, 11] and subspace detection [12]; on real data the same statistic localizes unplanted cyclone circulation. Open problems: identifiability *between* the diagonal and free-PSD excitation classes (e.g. banded or Toeplitz  $\mathbf{\Gamma}_S$  — where between reading off every weight and seeing only the image does the transition occur?); sharp first-order sparse thresholds over  $\ker \mathbf{B}_2$ ; effective sample sizes  $N_{\text{eff}}$  under temporal dependence; and tightening the Markov factor in (5) to match the empirical transition.

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## Compliance with ethical standards

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**Use of AI tools (disclosure per IEEE/SPS policy).** This preprint is an AI-assisted *draft*: substantial portions of the manuscript text, the source code, and the figures were generated with the assistance of an AI system (Claude, Anthropic) under the author’s direction across iterative adversarial-review cycles. Because IEEE/SPS policy does not permit AI-generated manuscript *text*, the author will independently rewrite the manuscript before any formal submission and then disclose only the permitted uses (editing, code, visualization, and review feedback). All mathematical statements are checked by the released test suite (49 tests, including exhaustive and Monte-Carlo checks for every quoted constant), and the full revision history is preserved unaltered in the public repository — git history is deliberately *not* rewritten to obscure AI use. The author takes full responsibility for the content.